Gen AI Assignment 1

QUESTION 2

Forward Pass:

$$\because \quad z1 = w1x + b1$$

$$= \begin{bmatrix} 0.42 & 0.72 \\ 0 & 0.3 \\ 0.15 & 0.09 \end{bmatrix} \begin{bmatrix} 0.34 \\ 0.55 \end{bmatrix}$$

$$z1 = \begin{bmatrix} 0.5388 \\ 0.165 \\ 0.1005 \end{bmatrix}$$

$$y1 = Relu(z1)$$

$$y1 = \begin{bmatrix} 0.5388 \\ 0.165 \\ 0.1005 \end{bmatrix}$$

$$z2 = w2y1$$

$$= \begin{bmatrix} 0.44 & 0.03 & 0.55 \end{bmatrix} \begin{bmatrix} 0.5388 \\ 0.165 \\ 0.1005 \end{bmatrix}$$

$$z2 = [0.297]$$

$$y2 = [0.297]$$

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$$z3 = w3y2$$

$$= \begin{bmatrix} 0.55 \\ 0.71 \\ 0.29 \end{bmatrix} \begin{bmatrix} 0.297 \end{bmatrix}$$

$$z3 = \begin{bmatrix} 0.163 \\ 0.21 \\ 0.086 \end{bmatrix}$$

$$y3 = \begin{bmatrix} 0.163 \\ 0.21 \\ 0.086 \end{bmatrix}$$

$$z4 = w4y3$$

$$= \begin{bmatrix} 0.97 & 0.55 & 0.97 \\ 0.71 & 0.7 & 0.22 \end{bmatrix} \begin{bmatrix} 0.163 \\ 0.21 \\ 0.086 \end{bmatrix}$$

$$z4 = \begin{bmatrix} 0.357 \\ 0.281 \end{bmatrix}$$

$$y4 = \begin{bmatrix} 0.357 \\ 0.281 \end{bmatrix}$$

$$J(\theta) = -0.34 \ln(0.357) - 0.55 \ln(0.281)$$

$$J(\theta) = 1.048$$

Backward Pass:

$$\nabla_{y4} Div = y^4 - actual$$

$$= \begin{bmatrix} 0.357 \\ 0.281 \end{bmatrix} - \begin{bmatrix} 0.34 \\ 0.55 \end{bmatrix}$$

$$= \begin{bmatrix} 0.017 \\ -0.269 \end{bmatrix}$$

$$\nabla_{z4} Div = \nabla_{z4} y4 \cdot \nabla_{y4} Div$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0.017 \\ -0.269 \end{bmatrix}$$

$$= \begin{bmatrix} 0.017 \\ -0.269 \end{bmatrix}$$

$$\nabla_{w4} Div = \nabla_{z4} Div \times y3^T$$

$$= \begin{bmatrix} 0.017 \\ -0.269 \end{bmatrix} \begin{bmatrix} 0.163 & 0.21 & 0.086 \end{bmatrix}$$

$$= \begin{bmatrix} 0.002 & 0.003 & 0.001 \\ -0.043 & -0.056 & -0.023 \end{bmatrix}$$

$$\nabla_{y3} Div = w4^T \times \nabla_{z4} Div$$

$$= \begin{bmatrix} 0.97 & 0.71 \\ 0.55 & 0.7 \\ 0.97 & 0.22 \end{bmatrix} \begin{bmatrix} 0.017 \\ -0.269 \end{bmatrix}$$

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$$\nabla_{y3}\ Div = \begin{bmatrix} -0.174 \\ -0.178 \\ -0.042 \end{bmatrix}$$

$$\nabla_{z3}\ Div = \nabla_{z3}\ y3 \times \nabla_{y3}\ Div$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -0.174 \\ -0.178 \\ -0.042 \end{bmatrix}$$

$$= \begin{bmatrix} -0.174 \\ -0.178 \\ -0.042 \end{bmatrix}$$

$$\nabla_{w3}\ Div = \nabla_{z3}\ Div \times y2^T$$

$$= \begin{bmatrix} -0.174 \\ -0.178 \\ -0.042 \end{bmatrix} \begin{bmatrix} 0.297 \end{bmatrix}$$

$$= \begin{bmatrix} -0.051 \\ -0.052 \\ -0.012 \end{bmatrix}$$

$$\nabla_{y2}\ Div = w3^T \times \nabla_{z3}\ Div$$

$$= \begin{bmatrix} 0.55 & 0.71 & 0.29 \end{bmatrix} \begin{bmatrix} -0.174 \\ -0.178 \\ -0.042 \end{bmatrix}$$

$$= \begin{bmatrix} -0.234 \end{bmatrix}$$

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$$\nabla_{z2} Div = \nabla_{z2} y2 \times \nabla_{y2} Div$$
$$= [1] [-0.234]$$
$$= [-0.234]$$

$$\nabla_{w2} Div = \nabla_{z2} Div \times y1^T$$
$$= [-0.234] [0.5388 \ 0.165 \ 0.1005]$$
$$= [-0.126 \ -0.038 \ -0.023]$$

$$\nabla_{y1} Div = w2^T \times \nabla_{z2} Div$$
$$= \begin{bmatrix} 0.44 \\ 0.03 \\ 0.55 \end{bmatrix} [-0.234]$$
$$= \begin{bmatrix} -0.102 \\ -0.007 \\ -0.128 \end{bmatrix}$$

$$\nabla_{z1} Div = \nabla_{z1} y1 \times \nabla_{y1} Div$$
$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -0.102 \\ -0.007 \\ -0.128 \end{bmatrix}$$
$$= \begin{bmatrix} -0.102 \\ -0.007 \\ -0.128 \end{bmatrix}$$

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$$\nabla_{w1} Div = \nabla_{z1} Div \times x^T$$

$$= \begin{bmatrix} -0.102 \\ -0.007 \\ -0.128 \end{bmatrix} [0.34 \quad 0.55]$$

$$= \begin{bmatrix} -0.034 & -0.056 \\ -0.002 & -0.003 \\ -0.043 & -0.07 \end{bmatrix}$$

New Weights:

$$w4 = \begin{bmatrix} 0.969 & 0.548 & 0.969 \\ 0.731 & 0.728 & 0.231 \end{bmatrix}$$

$$w3 = \begin{bmatrix} 0.575 \\ 0.736 \\ 0.296 \end{bmatrix}$$

$$w2 = [0.503 \quad 0.049 \quad 0.561]$$

$$w1 = \begin{bmatrix} 0.437 & 0.748 \\ 0.001 & 0.301 \\ 0.171 & 0.125 \end{bmatrix}$$

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QUESTION 3

Forward Pass:

$$z1 = \begin{bmatrix} 0.5388 \\ 0.165 \\ 0.1005 \end{bmatrix}$$

$$y1 = \begin{bmatrix} 0.5388 \\ 0.165 \\ 0.1005 \end{bmatrix}$$

~~$z2 = [0\ 29$~~ $u' = w_u \times y1$

$$u' = [0.5 \quad 0.5 \quad 0.5] \begin{bmatrix} 0.5388 \\ 0.165 \\ 0.1005 \end{bmatrix}$$

$$u' = [0.402]$$

$$u = [0.402]$$

$$\sigma' = w_\sigma \times y1$$

$$= [0.5 \quad 0.5 \quad 0.5] \begin{bmatrix} 0.5388 \\ 0.165 \\ 0.1005 \end{bmatrix}$$

$$\sigma' = [0.402]$$

$$\sigma = [0.402]$$

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$$z = \mu + \sigma \cdot \epsilon$$

Assuming $\epsilon = 1$ here for illustration but actually $\epsilon \sim N(0, 1)$.

$$z2 = [0.402] + [0.402][1]$$
$$z2 = [0.804]$$

$$z3 = w3\, z2$$
$$= \begin{bmatrix} 0.55 \\ 0.71 \\ 0.29 \end{bmatrix} [0.804]$$
$$z3 = \begin{bmatrix} 0.442 \\ 0.57 \\ 0.233 \end{bmatrix}$$
$$y3 = \begin{bmatrix} 0.442 \\ 0.57 \\ 0.233 \end{bmatrix}$$

$$z4 = w4\, y3$$
$$= \begin{bmatrix} 0.97 & 0.55 & 0.97 \\ 0.71 & 0.7 & 0.22 \end{bmatrix} \begin{bmatrix} 0.442 \\ 0.57 \\ 0.233 \end{bmatrix}$$

$$z4 = \begin{bmatrix} 0.968 \\ 0.764 \end{bmatrix}$$

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$$y^4 = \begin{bmatrix} 0.968 \\ 0.764 \end{bmatrix}$$

$$L_{recon} = -0.34 \ln(0.968) - 0.55 \ln(0.764)$$
$$= 0.159$$

$$KL = -\frac{1}{2}(1 + \log(\sigma^2) - \mu^2 - \sigma^2)$$
$$= -\frac{1}{2}(1 + \log(0.402^2) - 0.402^2 - 0.402^2)$$
$$= 0.572$$

$$J(\theta) = L_{recon} + KL$$
$$= 0.159 + 0.572$$
$$= 0.731$$

During backward pass, derivatives ∇_{y4} Div, ∇_{z4} Div, ∇_{w4} Div, ∇_{y3} Div, ∇_{z3} Div will be calculated in same way as done in autoencoder. After it ∇_{w3} Div will be calculated using:

$$\nabla_{w3} \text{Div} = \nabla_{z3} \text{Div} \times z2^T$$

Then; ∇_{z2} Div $= w3^T \times \nabla_{z3}$ Div

$$\nabla_u \text{Div} = 1 \times \nabla_{z2} \text{Div} + u$$

$$\nabla_{u'} \text{Div} = \nabla_{u'} u \times \nabla_u \text{Div}$$

$$= [1] \times \nabla_u \text{Div}$$

$$\nabla_o \text{Div} = \varepsilon \times \nabla_{z2} \text{Div} + (o - 1/o)$$

$$\nabla_{o'} \text{Div} = [1] \times \nabla_o \text{Div}$$

$$\nabla_{w_u} \text{Div} = \nabla_{u'} \text{Div} \times y1^T$$

$$\nabla_{w_o} \text{Div} = \nabla_{o'} \text{Div} \times y1^T$$

~~$\nabla_{y1} \text{Div} = w_u^T \times u' + w_o^T \times o'$~~

$$\nabla_{y1} \text{Div} = w_u^T \times \nabla_{u'} \text{Div} + w_o^T \times \nabla_{o'} \text{Div}$$

$$\nabla_{z1} \text{Div} = \nabla_{z1} y^1 \times \nabla_{y1} \text{Div}$$

$$\nabla_{w1} \text{Div} = \nabla_{z1} \text{Div} \times x^T$$

So, in the end ∇_{z1} Div and ∇_{w1} Div will also be calculated similarly as done in backward pass in autoencoder. In this way, we will calculate all weight change matrices and update the weights.